

Field and Galois Theory

Appetizer

1.

- (a) Give an example of a tower of field extensions $F \supset L \supset K$ such that L/K is Galois and F/L is Galois, but F/K is not Galois.
- (b) Give an example of a primitive element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Prove that the element you give is indeed a primitive element. Find the minimal polynomial of the element you find.

* 2.

Let $\mathbb{F}_p$ denote the field with p elements.

- (a) What is the factorization of the polynomial $x^4 + 1$ into irreducible factors in $\mathbb{F}_3[x]$?
- (b) Give an example of an element $a \in \mathbb{F}_5$ such that $x^4 - a$ is irreducible in $\mathbb{F}_5[x]$, and prove that $x^4 - a$ is irreducible with this choice of a .
- (c) Prove that for any nonzero choice of $a \in \mathbb{F}_{103}$, the polynomial $x^4 - a$ factors either as a product of two irreducible quadratic polynomials, or as a product of two linear polynomials and an irreducible quadratic.

3.

Let ζ be a primitive k th root of unity.

- (a) Prove that $K = \mathbb{Q}(\zeta)$ is a Galois extension of $\mathbb{Q}$.
- (b) Compute $\text{Gal}(K/\mathbb{Q})$.

Entrée

* 4.

- (a) Let a be a root of the polynomial $x^4 - 7 \in \mathbb{Q}[x]$. Show that $\mathbb{Q}(a)$ is not Galois over $\mathbb{Q}$.
- (b) Let F be a field of characteristic 5, and let b be a root of $x^4 - 7 \in F[x]$. Show that $F(b)$ is Galois over F .
- (c) Compute the Galois group $\text{Gal}(F(b)/F)$.

* 5.

This is a question about “biquadratic extensions” in two parts.

- (a) Let F/E be a degree-4 Galois extension, where E and F are fields of characteristic different from 2. Show that $\text{Gal}(F/E) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ if and only if there exist $x, y \in E$ such that $F = E(\sqrt{x}, \sqrt{y})$ and none of x, y, xy are squares in E .

- (b) (Slightly modified from original) Give an example of an infinite field E of characteristic 2 that is not algebraically closed but that has no Galois extension F/E with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$

6.

Suppose that $K/\mathbb{Q}$ is a degree-4 Galois extension with Galois group $\mathbb{Z}/4\mathbb{Z}$. Prove that $\sqrt{-1} \notin K$.

* **7.**

For each of the following statements, either prove it or provide a counterexample.

- (a) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{C})$ is diagonalizable.
- (b) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{Q})$ is diagonalizable.
- (c) Every matrix in $\mathrm{GL}_n(\overline{\mathbb{F}_p})$ has finite order. (We write $\overline{\mathbb{F}_p}$ for the algebraic closure of the finite field $\mathbb{F}_p$.)
- (d) Every matrix of finite order in $\mathrm{GL}_n(\overline{\mathbb{F}_p})$ is diagonalizable.

* **8.**

Let L/K be a degree p Galois extension, where p is prime, and let σ be a nontrivial element of $\mathrm{Gal}(L/K)$. Then σ is a linear transformation of the K -vector space L .

- (a) What is the characteristic polynomial of σ ?
- (b) Write L^0 for the subspace of L consisting of elements with trace 0. Show that σ sends L^0 to L^0 .
- (c) What is the characteristic polynomial of σ acting on L^0 ?